Tool-call gains did not survive NF4

Pernav Jain | Soup AI Engineer take-home | 10 November 2026
Run `20261110T044014005029Z` | **DON'T SHIP**

The dense merge improves exact tool-call correctness from 50.0% to 60.0%. The saved NF4 model scores 49.3%, loses 22 previously correct turns and gains six. Its general tool-call score also regresses. The recorded merge-equivalence gate remains unresolved. Training and export completed, but this 4-bit artifact fails the release criteria.

1. Experiment and memory

Qwen2.5-1.5B-Instruct fits the model-size requirement and supports a tractable English task: emit ordered JSON tool calls. Hermes supplies tool schemas and reference calls. The prepared split contains 1,600 train, 150 validation and 150 test turns, with tool/request families separated. Every row fits 1,024 tokens; assistant content and EOT receive loss. Structural audits cover all 1,900 rows. References were not exhaustively adjudicated for semantic correctness.

Soup `0.77.2` at commit `4cff43bf967e` trains a frozen FP16 base with LoRA on seven projections per layer: rank 8, alpha 16, dropout 0.05; AdamW, LR `1e-4`, cosine decay, 5% warmup, batch 1 and accumulation 8. Each of seeds 42/43/44 completes two epochs, or 400 steps. Epoch validation loss selects the checkpoint; validation exact correctness selects seed 42. Test data select neither. One T4 is exposed from Kaggle's two-card allocation; usable capacity is 14.562 GiB.

The preserved pre-training estimate contains the following floors and a **12 GiB planning envelope**:

Component	Calculation	GiB
Frozen weights	1,543,714,304 x 2 bytes	2.875
LoRA, gradients, Adam moments	9,232,384 x (4 + 4 + 8) bytes	0.138
One full logits tensor	1 x 1,024 x 151,936 x 2 / 4 bytes	0.290 / 0.580
NF4 storage floor	1,310,195,712 x 0.5 + 233,373,696 x 2 bytes	1.045

Checkpointed activations, loss workspaces, allocator/context overhead and export staging sit outside these floors. A retained FP16 hidden state per layer alone is about 0.082 GiB; this reconstruction is a lower bound, not a full activation forecast. The original activation allowance was coarse.

Measured phase	GPU allocated peak	Sampled device peak	Host RSS peak
Training, maximum across seeds	4.855 GiB	5.569 GiB	3.884 GiB
Dense merge	0.034 GiB	0.138 GiB	6.436 GiB
Dense staging and NF4 save	1.089 GiB	1.247 GiB	6.132 GiB

Global allocator reservation peaks at 5.430 GiB, leaving 8.993 GiB of measured headroom. The allocated peak exceeds the base-plus-state floor by 1.842 GiB; that residual includes activations and workspaces. Two dense copies would require 5.751 GiB of host storage, consistent with the order of the export RSS, but tensor-level host attribution was not measured. CUDA counters and 0.2-second-target device/RSS samples describe different quantities; samples can miss spikes and independent peaks cannot be added. M/Q weight files occupy 2.875/1.065 GiB: 2.70x compression, limited by floating-point embeddings and quantization metadata.

2. Evidence that the adapter trained

For seed 42, all 392 LoRA tensors change; they contain **9,232,384 scalar weights**. Optimizer membership matches trainable tensors. There are 399 nonzero-LR optimizer steps, one zero-LR warmup step and no AMP skips; the first materialized update is `5.01e-6`. Selected frozen projection hashes remain unchanged. Disabling the adapter restores the base's probe log probabilities exactly; enabling it changes them by up to 18.47. These probes cover four validation rows, 416 reference tokens and 16 distribution positions.

The initial zero-effect adapter is rejected by Soup's adapter check. Substituting the base for M fails the weight test; substituting quantized base Q0 for Q fails identity checks. Cold M/Q reloads block reads from the base/adapter folders. Q contains 196 NF4 linear modules, nested quantization state and FP16 compute. These checks establish an active, changed adapter and real exports; they do not establish desirable behavior or universal equivalence. The training loss of 0.7271 to 0.0023 alone would establish neither.

3. Held-out artifact comparison

Complete correctness requires the reference's tool names, JSON argument values, types, call count and order. Invalid or length-stopped responses fail. All 150 test turns remain in the denominator. B is the base, M the dense merge and Q the saved NF4 model.

Artifact	Correct turns	Exact correctness	Parseable	Assistant NLL
B	75 / 150	50.00%	94.00%	0.64338
M	90 / 150	60.00%	98.00%	0.04658
Q	74 / 150	49.33%	94.67%	0.05657

Paired bootstrap over 150 held-out groups, 10,000 resamples: **M-B +10.00 pp [2.00, 18.00]**; **Q-M -10.67 pp [-17.33, -4.00]**; **Q-B -0.67 pp [-7.33, 6.00]** (95% intervals). The eight bundled general suites include a tool-call regression of 70.0% to 57.5% to 55.0% for B/M/Q. No new critical failure appears on six fixed challenge turns; this small set supports no broad safety claim. Across seeds, validation exact correctness is 56.67%, 54.67% and 54.67%; the final test evaluates only selected seed 42.

4. Silent failures and check coverage

Boundary	Observed check and result
Data and scoring	Strict parser/schema fixtures reject wrong arguments, duplicate keys, type confusion and extra/reordered calls. Soup format validation accepts the malformed-call fixture; it does not inspect that payload's semantics.
Tokenizer/template	Exact response/EOT masks and padding checks pass on every row. Soup doctor rejects the length-boundary fixture (exit 2) and accepts the missing-tools fixture (exit 0). Format validation accepts the double-template fixture (exit 0).
Training	Fingerprints, actual updates, adapter on/off probes and frozen-weight hashes corroborate training. Soup catches the initialized no-op adapter (exit 2); a falling loss is insufficient evidence.
Merge	Four sampled matrices equal FP16-rounded W + scaled BA, with zero residual. However, A/M maximum reference-logprob error is 0.03867, above the frozen 0.02000 limit; mean KL is 1.08e-6. This is an unresolved numerical-equivalence failure, not proof of a wrong export.
Quantization	Q is a standalone NF4 checkpoint, but loses 10.67 pp against M. Quantization metadata and valid JSON do not guarantee retained task behavior.
Inference/release	Cold-load identity/device checks pass. Soup `ship --evidence` returns DON'T SHIP for both M and Q using saved measured scores; this is offline decision replay, not another model evaluation.

The holdout distribution differs from training: Glaive supplies 1,081/1,600 training turns, while primary Hermes supplies 148/150 test turns. Exact-reference grading can reject alternative valid calls. Four projection checks and four probe rows cannot establish whole-model merge equivalence. Eight small general suites and one selected-seed test leave substantial uncertainty.

5. Verdict and required changes

DON'T SHIP. Six frozen gates fail: merge fidelity, minimum Q/B gain, positive gain interval, maximum Q/M loss, its confidence bound and general-suite retention. Soup independently applies its decision rules to the supplied measured scores and also rejects both candidates.

First investigate A/M differences with matched precision and kernels and a wider validation probe set; retain the failed 0.02 gate rather than relax it after seeing the result. Then inspect the 22 M-correct/Q-wrong turns and compare revised export settings on validation. Address general tool-call retention with better development coverage and data balance. Any tuned successor needs a fresh held-out evaluation before release; the current test has been inspected.

Reflection and AI use

The surprise was how much exact correctness fell after NF4 despite the small NLL increase from 0.0466 to 0.0566. I remain concerned about semantic reference quality and the limited merge probes. I set the fine-tuning settings in the Soup configuration, accepted the three-seed evaluation and rejected a LoRA-versus-QLoRA training comparison because it was outside the requested pipeline. AI assisted the notebooks, instrumentation, debugging and report drafting; the submission's CPU review checks its measurements against the preserved run. My next step would be to investigate export fidelity and quantization errors before further training.

Evidence: `run/verdict.json`, `run/evidence/{training-42,fidelity-and-export-controls,memory-estimate-before}.json`; full mapping in `docs/evidence-map.md`. Repository: https://github.com/here-2007/soup-hermes-t4-takehome. Exact weights: https://www.kaggle.com/models/pernavjain/soup-experiment/PyTorch/experiment-artifacts/1.